

Power Distribution System Planning for Mitigating Overheating Risk Inequity

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Abstract—Extreme heat events are increasingly impacting the operation of power distribution systems, causing overloads due to excessive use of air conditioning systems, forcing power utilities to take measures to avoid large-scale blackouts in a bulk transmission system, including rotating outages. However, these power outages expose communities to overheating risk as air conditioning ceases operation, which is inequitably distributed and affects vulnerable communities mainly. This paper proposes a framework to quantify and mitigate the outage-induced overheating risk during a heat wave, which corresponds to the overheating that can be avoided by utility action. The framework features a metric that captures the impact of power outages during heat waves on individual buildings, which is aggregated to derive system-level metrics to assess the resilience of a distribution grid. This metric is integrated into a distribution planning model to drive decisions on distribution grid investments in an equity-informed manner. The framework is evaluated in modified IEEE 33-node and 123-node test systems, where the ability of the framework to capture and equitably allocate distributed resources is validated, paving the way for regulatory frameworks that account for social equity impact in the power sector.

Index Terms—Distributed energy resources, energy justice, power distribution planning, resource adequacy, rotating outages.

NOMENCLATURE

Sets and Indices

c	Index of building archetypes.
t	Index of time steps.
a	Index of feeder sections.
i	Index of nodes.
s	Index of automated switches.
$\mathcal{T}$	Set of time steps in the planning horizon.
$\mathcal{T}'$	Subset of time steps in the overheating evaluation horizon.
$\mathcal{T}^*$	Subset of time steps where the heat wave occurs.
$\mathcal{C}$	Set of building archetypes.

$\mathcal{A}$	Set of feeder sections.
$\mathcal{B}_a$	Set of nodes in feeder section a .
$\mathcal{B}'$	Set of candidate nodes.
$\mathcal{S}$	Set of automated switches.
$\mathcal{L}$	Set of line sections.
<i>Parameters</i>	
Δt	Time step.
θ^{th}	Caution temperature threshold.
D	Outage duration.
R_c, C_c	Thermal resistance/reactance of building with archetype c .
θ_t^{out}	Outdoor air temperature at time t .
$\theta_{c,t}^{ref}$	Indoor temperature setpoint for building c , time t .
COP_c	Coefficient of performance of the HVAC system at building c .
$\hat{t}$	Outage starting time.
$N_{i,c}$	Number of buildings c connected at node i .
A_{pv}, A_{es}	Coefficient to annualize solar unit/energy storage investments.
c^{fg}, c^{fe}	Solar unit/energy storage fixed installation cost.
cv^g	Solar unit variable investment cost per kW.
cv^e	Energy storage unit variable investment cost per additional kW output.
cv^s	Energy storage unit variable investment cost per additional kWh capacity.
of^g	Solar unit annual fixed maintenance cost per kW.
of^e	Energy storage unit annual fixed maintenance cost per additional kW output.
of^s	Energy storage unit annual fixed maintenance cost per additional kWh capacity.
ov^g, ov^e	Solar unit/Energy storage operation unit cost per kW.
λ_t^{grid}	Electricity price at time t .
c_i	Outage cost at node i .
$p_{i,t}^l, q_{i,t}^l$	Active/reactive power demand at node i , time t .
$1/N$	Number of rotating outage events during a heat wave per year.
$\bar{t}$	Power supply restoration time.
$\bar{B}$	Available investment budget.
p_c^{HVAC}	Maximum power capacity of the HVAC system at building c .
$s_{ij,t}$	Binary parameter for the status of the automated switch at line ij , time t .
r_{ij}, x_{ij}	Resistance and reactance of line ij .

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M	Large positive real number.
$\bar{v}, v$	Maximum/minimum squared voltage limit.
$\bar{p}^g$	Maximum allowed solar capacity per node.
$\bar{q}^g$	Maximum reactive power supplied by the inverter.
$\bar{p}^e$	Maximum allowed storage power output per node.
$\bar{E}$	Maximum allowed storage capacity per node.
r_t^{solar}	Normalized solar radiation at time t .
η_i^c, η_i^d	Charging/discharging efficiency of energy storage system at node i .
$\beta_{i,t}$	Binary variable that enables the charging operation.
Variables	
$OH_{c,\hat{t},D}$	Outage induced overheating risk in building c for an outage starting at time $\hat{t}$ with duration D .
$TH_{c,\hat{t},D}$	Total overheating risk in building c for an outage starting at time $\hat{t}$ with duration D .
$NH_{c,\hat{t},D}$	Non-outage induced overheating risk in building c for an outage starting at time $\hat{t}$ with duration D .
$\theta_{c,t}$	Indoor air temperature at building c at time t .
$\theta_{c,t}^o$	Indoor air temperature with no outage at building c at time t .
$p_{c,t}^{HVAC}$	Power consumed by the HVAC system at building c , time t .
$\Psi_{a,\hat{t},D}$	Total overheating risk in feeder section a for an outage starting at $\hat{t}$ with a duration D .
$\Pi_{a,\hat{t},D}$	Total outage-induced overheating risk in feeder section a for an outage starting at $\hat{t}$ with a duration D .
$\phi_{a,\hat{t},D}$	Outage-induced overheating risk per kWh in feeder section a for an outage starting at $\hat{t}$ with a duration D .
$\Omega_{a,\hat{t},D}$	Outage-induced overheating risk per consumer in feeder section a for an outage starting at $\hat{t}$ with a duration D .
$p_{i,t}^{l*}$	Power curtailed from node i , time t .
α_i^g	Binary variable to decide if a solar unit is installed at node i .
α_i^e	Binary variable to decide if an energy storage is installed at node i .
$p_i^{g,new}$	Added output for the solar unit at node i .
$p_i^{e,new}$	Added output for the energy storage unit at node i .
E_i^{new}	Added capacity for the energy storage unit at node i .
$p_{i,t}^g, q_{i,t}^g$	Active/reactive power generated by solar unit at node i , time t .
$p_{i,t}^c, p_{i,t}^d$	Charging/discharging power for energy storage unit at node i , time t .
$q_{i,t}^e$	Reactive power injection from energy storage unit at node i , time t .
$p_{i,t}^{grid}, q_{i,t}^{grid}$	Active/reactive power procured from the bulk transmission system at node i , time t .
$e_{i,t}$	Energization status of node i , time t .
$\bar{\Phi}$	Maximum overheating risk per kWh among the power distribution system sections.

$\tilde{\phi}_{a,\hat{t},D}$	Outage-induced overheating risk per kWh in feeder section a for an outage starting at $\hat{t}$ with a duration D , calculated as a function of the section energization status.
$\hat{e}_{a,t}$	Energization status of feeder section a at time t .
$v_{i,t}$	Squared voltage at node i , time t .
$E_{i,t}$	Energy stored in energy storage system i , time t .

I. INTRODUCTION

EXTREME weather events induced by climate change are increasingly compromising the ability of power distribution systems to maintain a reliable electricity supply during such occurrences. More specifically, heat waves, which are becoming more frequent and intense in Western US [1], lead consumers to excessively use Heating, Ventilation, and Air Conditioning (HVAC) systems to maintain safe and comfortable indoor temperatures in buildings. The massive HVAC utilization causes overloads on the electric distribution side which, combined with insufficient power generation and limited power transfer capability on the transmission side—also affected by the excessive heat—, compromise the generation-demand balance and increase the risk of large-scale blackouts in the bulk transmission system [2].

When the bulk transmission system is at risk of a widespread blackout during a heat wave, Independent System Operators urge power utilities to implement measures to alleviate overloading. These measures include a last-resort strategy of applying rotating outages to sequentially and alternatively disconnect power distribution circuits from the power substations during the peak demand. While rotating outages avoid the risk of large-scale blackouts, these maneuvers disconnect consumers for a few hours during a heat wave and expose communities to the adverse effects of heat. The impact of the heat wave, however, is not equally distributed among the community members affected by a power outage, as some community members may experience fast temperature spikes due to inadequate house insulation, posing a higher risk to populations with heat-sensitive groups, such as children and older adults [3], [4], [5]. For instance, a study presented in [6] discusses the risk of power grid failures during heat waves for 2.8 million residents across Atlanta, GA, Detroit, MI, and Phoenix, AZ.

The aftermath of controlled rotating outages during extreme heat can be mitigated by equipping the power distribution circuit with distributed energy resources (DERs), e.g., solar generation or energy storage (ES) systems, capable of sustaining the power supply for groups of islanded nodes when a rotating outage is imposed at the substation. However, resilient DER investment decisions in power distribution systems are informed exclusively by techno-economic and infrastructure criteria, aiming at reducing overall outage costs. These criteria tend to place DERs closer to the high consumption points, often located in higher-income neighborhoods, neglecting the impact of outages on the population and their asymmetries across sociodemographic groups [7], such that disadvantaged communities experience longer outage duration [8], have limited access to rooftop solar [9], [10],

[11], are connected to distribution grid nodes with less hosting capacity [12], and in general perceive a higher hardship during outages [13].

A. Literature Review

The impact of heat waves on power system operation and planning is studied in the literature. A planning model that considers the impact of heat waves and extreme droughts on the performance of the power generation units and the power system load is presented in [14]. A method to prioritize upgrades in power distribution system combining Shortest Path Problem and Markov chain modeling for enhancing resilience to heat waves is presented in [15]. A strategy for optimal DER dispatch in distribution systems exposed to a heat wave is formulated in [16], considering the temperature-imposed generation limits. A model to address the generation derating induced by excessive temperature is presented in [17] to assess the impact of heat waves in the unit commitment problem.

Bringing equity into power distribution system planning for resilience enhancement has gained momentum given the disparities perceived when communities are exposed to high-impact-low-probability events, such as extreme weather [18]. Planning for resilience has commonly focused on enhancing the power grid ability to withstand natural disasters that involve infrastructural damage that may compromise consumers and other critical infrastructures [19], through the utilization of distributed energy resources [20], distribution line hardening, and automation operation [21]. A generation and transmission expansion strategy is proposed in [22] to consider seismic risk in the formulation. Other approaches consider enhancing resilience against wind-induced climatic hazard. A resilience-oriented design is proposed in [23] to harden lines, install distributed generation and automated switches. Resilience enhancement is achieved in [24] by using mobile energy storage to create dynamic microgrids. A tri-level hardening plan to enhance resilience considering reconfiguration and distributed generation is proposed in [25]. The conditional value at risk is considered in [26] to plan line and storage device investment while balancing between routine and extreme events. A robust optimization approach for hardening and DG placement against natural disasters is formulated in [27] considering natural disaster uncertainty. Similarly, a two-stage stochastic approach is used in [28] for the installation of switches, tie lines, and line hardening. A two-stage stochastic optimization is proposed in [29], which uses k -means to reduce the number of scenarios evaluated. Alternatively, a three-stage stochastic model to enhance resilience is proposed considering line damage and load uncertainty in [30]. A resilience-oriented design is proposed in [31] for topological and structural upgrades to reduce the system susceptibility to extreme weather.

The conventional approaches to enhance power distribution system resilience prioritize the economic benefits while disregarding the vulnerability of communities to extreme weather. Also, planning decisions have mostly focused on extreme weather events that cause damage to the infrastructure itself, while some events such as heat waves, which affect the operation and expose power systems to large-scale blackouts have not been

addressed. The effects of excessive heat on communities have been assessed and quantified (e.g., [32], [33]), but their inclusion into distribution system planning is not studied in the literature.

B. Contribution and Paper Structure

Considering the disruptive nature of heat waves in the power grid operation and the consumer risk of overheating, this paper proposes a framework for the siting and sizing of distributed energy resources on a distribution grid with explicit equity criteria, where the overheating risk of consumers throughout the power distribution system informs the prioritization of distribution grid upgrades. While several external events, such as wildfires and vegetation, may cause power outages that expose communities to overheating, this paper is focused on the occurrence of power outages during heat waves. First, a set of metrics is formulated and assessed to categorize vulnerable feeder sections where investments in DERs lead to reductions in the overheating risk induced by a power outage when added. The properties of these metrics are discussed to define the proper use in a power systems context, in particular for equity-aware distribution grid planning. Then, an optimization model is formulated to integrate the overheating risk induced by power outages during heat waves into the power distribution system planning. The contributions of this paper are:

- An overheating risk metric to be applied in the context of power distribution is proposed. This metric can be used to inform decisions around power outages during heat waves, and capture their consequent impact on vulnerable populations.
- The concept of *outage-induced overheating risk* is introduced to capture the portion of the overheating experienced by the consumer that can be avoided by utility's decisions. Such concept is instrumental to formalize regulatory and engineering procedures for distribution grid resilience against heat waves.
- A discussion on how individual household overheating metrics can be aggregated to derive system-level metrics is presented, which can be used to assess performance of an entire distribution grid regarding resilience in the context of heat waves.
- We show how the proposed metric framework can be used to drive distribution grid investments in an equity-informed distribution planning model.
- We show that grid investment planning with cost and equity criteria result in different solutions. These results not only demonstrate the relevance of the metric framework proposed, but also pave the way for new regulatory frameworks and utility planning methodologies account for social equity impacts of decisions in the power sector.

The remainder of the paper is organized as follows: The calculation of the overheating risk caused by a power outage during a heat wave and its mapping into the power distribution system is presented in Section II. The optimization models for equity-aware power distribution planning are formulated in Section III. The metrics and optimization models are evaluated in Section IV and conclusions are presented in Section V.

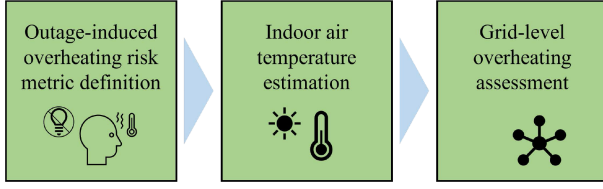


Fig. 1. Framework for outage-induced overheating risk evaluation.

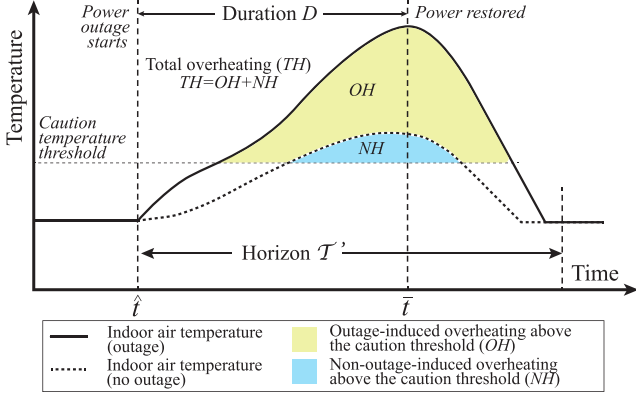


Fig. 2. Overheating risk induced by power outage during a heat wave.

II. DETERMINING OUTAGE-INDUCED OVERHEATING RISK IN POWER DISTRIBUTION SYSTEMS

In this section, the overheating risk caused by a power outage during a heat wave is illustrated and a framework for its calculation and mapping to the power distribution system is introduced, as summarized in Fig. 1. The proposed framework first defines and quantifies the outage-induced overheating risk. The overheating risk assessment relies on defining whether the household in a building experiences indoor air temperatures above the risk threshold, hence a method to estimate the indoor air temperature is presented. Finally, the metrics are evaluated based on the temperature assessment and mapped into the power distribution system to analyze the concentration of overheating-exposed vulnerable communities.

A. Effect of Power Outage in the Household Overheating Risk

The exposure of a community to adverse effects of extreme heat is intensified when combined with a power outage event, which can significantly affect the indoor air temperature perceived inside households. The impact of the power outage on the building indoor air temperature is illustrated in Fig. 2. Let us assume that the HVAC system at a building is operating normally and the temperature remains fixed in a temperature setpoint. In a no-power-outage scenario, the temperature response follows the trajectory shown by the dashed line. The building indoor air temperature may deviate from its setpoint if the HVAC maximum cooling capacity is reached and may exceed the caution temperature threshold. In a power outage, the temperature quickly deviates from the setpoint due to the lack of an operational HVAC system. In this outage scenario, the temperature may exceed the

caution temperature threshold faster and increase the risk of overheating. The rate at which temperature varies depends on the building envelope, as poorly insulated buildings with higher heat gains experience faster temperature changes than buildings with better thermal properties. Also, the temperature deviation induced by the outage depends on its time and duration.

The overheating risk of a household is calculated based on a thermal assessment to evaluate the indoor air temperature during a heat wave [33]. The household overheating can be described as shown in Fig. 2, which is determined from the area below the temperature and the caution temperature threshold [34]. As previously mentioned, a household can be exposed to indoor air temperatures that exceed the caution threshold even in a no-power-outage scenario due to the building envelope and the cooling system efficiency, which is represented by the blue area in Fig. 2. Hence, from a power system application perspective, it is important to distinguish between the overheating component that is caused by the outage from the overheating level related to the building itself, that occurs even without any power disconnection. The total overheating risk in a building c after a power outage occurring at time $\hat{t}$ with a duration D , denoted as $TH_{c,\hat{t},D}$, corresponds to the combination of the outage-induced overheating risk shown in the yellow area, $OH_{c,\hat{t},D}$, and the overheating not related to the power outage shown in the blue area, $NH_{c,\hat{t},D}$, depicted in Fig. 2. The outage-induced overheating risk can be interpreted as the overheating that can be avoided or reduced if the building electricity supply is sustained throughout the rotating outage. In other words, it is the portion of the overheating experienced by the consumer that can be avoided by utility's actions. Mathematically, these components can be evaluated in a horizon T' from the outage occurrence as:

$$OH_{c,\hat{t},D} = TH_{c,\hat{t},D} - NH_{c,\hat{t},D}, \quad (1)$$

$$TH_{c,\hat{t},D} = \sum_{t \in T'} \max\{\theta_{c,t} - \theta^{th}, 0\} \Delta t, \quad (2)$$

$$NH_{c,\hat{t},D} = \sum_{t \in T'} \max\{\theta_{c,t}^o - \theta^{th}, 0\} \Delta t, \quad (3)$$

where $\theta_{c,t}$ is the indoor air temperature during the outage at building c at time t , θ^{th} is the caution temperature threshold, $\theta_{c,t}^o$ is the indoor air temperature with no outage at building c , time t , and Δt is the time step between samples. It is worth noting that the overheating calculation presented here determines risk based on household exposure to temperatures above a caution threshold. Other aspects that intensify heat vulnerability, such as the presence of children or older adults, may also be considered in the overheating risk evaluation.

B. Indoor Air Temperature Estimation

The outage-induced overheating risk is calculated based on the indoor air temperature of the buildings during the heat wave, which depends on the building envelope. For this assessment, it is assumed that buildings are modeled using a first-order RC model approach [35]. The indoor air temperature for a household c at time t , $\theta_{c,t}$, can be calculated using the state equation:

$$\theta_{c,t} = a_c \theta_{c,t-1} - b_c COP_{c,t-1}^{HVAC} + d_c \theta_{t-1}^{out}, \quad (4)$$

where $p_{c,t}^{HVAC}$ is the power consumption of HVAC system of household c at time t , COP_c is the coefficient of performance of the HVAC system installed at c , θ_t^{out} is the outdoor air temperature, and a_c, b_c, d_c are coefficients related to the building envelope, calculated as a function of the building thermal resistance R_c and thermal capacitance C_c as $a_c = 1 - \Delta t / R_c C_c$, $b_c = \Delta t / C_c$, and $d_c = \Delta t / R_c C_c$. The power required for HVAC operation is a function of the difference between the outdoor air temperature and the indoor temperature setpoint for building c at time t , $\theta_{c,t}^{ref}$, obtained as:

$$p_{c,t}^{HVAC} = \max \left\{ \min \left\{ \frac{\theta_t^{out} - \theta_{c,t}^{ref}}{COP_c R_c}, \overline{p_c^{HVAC}} \right\}, 0 \right\}, \quad (5)$$

where $\overline{p_c^{HVAC}}$ is the maximum power capacity of the HVAC system installed at building c . An HVAC system can maintain the indoor air temperature at the defined setpoint as long as it operates within its maximum power capacity (5). For the thermal simulation, it is assumed that the indoor air temperature at time t is equal to the temperature setpoint if the HVAC power consumption is less than its maximum limit, and the difference between the temperature at time $t - 1$ and the temperature setpoint is no more than 1 °C. It is worth noting that the setpoint influences the temperature trajectory calculated with (4) during the power outage, as buildings with a lower setpoint experience less temperature increase during the power outage. However, lower setpoints lead to higher energy consumption, which is undesirable as it increases the risk of a rotating outage and does not represent a benefit if the cooling system has reached its maximum cooling capacity. Regardless of the temperature setpoint, the overheating risk can still be calculated with the metrics introduced in (1)–(3), as the overheating risk under a different setpoint is determined based on the exposure to temperatures above the caution.

C. Overheating Risk Metrics for Distribution Grids

Let $\mathcal{A} = \{1, \dots, A\}$ be the set of feeder sections of the distribution system that can remain energized during a heat wave. Assuming that these sections have different number of consumers, and they serve populations with heterogeneous vulnerability characteristics, it is important to derive a metric that describes the overall overheating risk in a feeder section. Four options are considered to derive this system-level metric.

1) *Total Overheating Risk* (Ψ): This metric corresponds to the aggregation of the overheating perceived in the feeder section a for an outage starting at time $\hat{t}$ with a duration D , calculated as:

$$\Psi_{a,\hat{t},D} = \sum_{i \in \mathcal{B}_a} \sum_{c \in \mathcal{C}} N_{i,c} TH_{c,\hat{t},D}, \quad \forall a \in \mathcal{A}, \quad (6)$$

where $N_{i,c}$ is the number of buildings with archetype c connected to node i , $\mathcal{B}_a$ is the subset of nodes contained in feeder section a , and $\mathcal{C}$ is the set of building archetypes. This metric combines both the outage- and non-outage-induced overheating, hence it helps quantify the total exposure and risk of consumers to overheating during the heat wave.

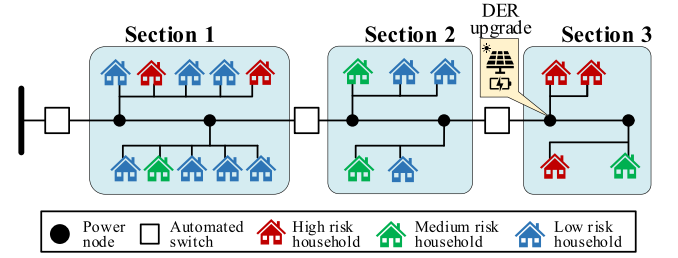


Fig. 3. Example of resulting distribution system sections during the rotating outage. DERs installed at nodes supply power to all consumers within the section.

2) *Total Outage-Induced Overheating Risk* (Π): This metric corresponds to the total aggregation of the overheating risk from all the buildings being supplied at feeder section a for an outage starting at time $\hat{t}$ with a duration D , calculated as:

$$\Pi_{a,\hat{t},D} = \sum_{i \in \mathcal{B}_a} \sum_{c \in \mathcal{C}} N_{i,c} OH_{c,\hat{t},D} \quad \forall a \in \mathcal{A}. \quad (7)$$

This metric aggregates the total overheating in the feeder section and represents the total vulnerability that is reduced if the section remains energized.

3) *Outage-Induced Overheating Risk per kWh* (ϕ): This metric relates the overheating risk in the feeder section with the energy curtailed during the evaluation period $\mathcal{T}'$:

$$\phi_{a,\hat{t},D} = \frac{\sum_{i \in \mathcal{B}_a} \sum_{c \in \mathcal{C}} N_{i,c} OH_{c,\hat{t},D}}{\sum_{i \in \mathcal{B}_a} \sum_{t \in \mathcal{T}'} p_{i,t}^{l*} \Delta t}, \quad \forall a \in \mathcal{A}, \quad (8)$$

where $p_{i,t}^{l*}$ is the power curtailed from node i , time t . This metric represents the overheating risk intensity, which increases with higher presence of high-risk households. In other words, it captures the overheating reduction potential of each kWh served by the utility.

4) *Outage-Induced Overheating Risk per Consumer* (Ω): This metric normalizes the overheating risk in the feeder section with respect to the total number of consumers and assigns a normalized overheating risk measure per supplied household in the feeder section:

$$\Omega_{a,\hat{t},D} = \frac{\sum_{i \in \mathcal{B}_a} \sum_{c \in \mathcal{C}} N_{i,c} OH_{c,\hat{t},D}}{\sum_{i \in \mathcal{B}_a} \sum_{c \in \mathcal{C}} N_{i,c}}, \quad \forall a \in \mathcal{A}. \quad (9)$$

This metric provides a sensitivity representing the community in the feeder section as an equivalent consumer, where higher presence of vulnerable households leads to a larger overheating per consumer.

D. Metric Interpretation

To illustrate the insights obtained from each metric, consider the distribution test feeder shown in Fig. 3. The test feeder is split in three sections given the automated switch locations and three building archetypes are considered with different overheating risk—high, medium, low risk—with a 4 kW load per household. The overheating risk is calculated as described in Section II-A for outage duration of 4 and 2 hours and summarized in Table I. The evaluation of the metrics (6)–(9) for the

TABLE I
OVERHEATING (OH) PER HOUSEHOLD

	$D = 4$ hours		$D = 2$ hours	
Household type	Total OH (°C-h)	Outage-induced OH (°C-h)	Total OH (°C-h)	Outage-induced OH (°C-h)
High risk	19.19	18.13	7.13	6.071
Med. risk	10.58	10.58	1.13	1.13
Low risk	5.96	5.96	0.181	0.181

TABLE II
OBTAINED METRICS FOR THE TEST EXAMPLE FOR A 4-HOUR OUTAGE

Section	Metric Ψ (°C-h)	Metric Π (°C-h)	Metric ϕ (°C-h/kWh)	Metric Ω (°C-h/consumer)
1	90.65	88.53	0.632	8.852
2	39.03	39.03	0.558	7.807
3	68.15	64.96	1.160	16.24

configuration shown in Fig. 3 for a power outage with a duration of 4 hours is summarized in Table II, where all automated switches are tripped to split the test feeder in sections.

The results in Table II show different insights on the overheating risk concentration and intensity throughout the power distribution system. Feeder section 1 obtains the highest value of metrics Ψ and Π corresponding to the largest accumulation of overheating risk, caused by the high number of consumers compared to the other sections. Metric Ψ is higher than Π due to consumers in the feeder section with overheating that occurs even without power outage due to the building envelope. The overheating risk intensity is captured with metric ϕ ; by normalizing the metric based on the load, the metric assumes higher values in feeder sections where the concentration of high-risk consumers is greater compared to the installed load in the section. According to metric ϕ , feeder section 2 is the most critical as it has the largest concentration of consumers with higher overheating risk, followed by section 1 and last section 2, which has no high-risk consumer. Metric Ω has a similar behavior to metric ϕ , assigning a normalized overheating risk to each consumer based on the number of consumers within the feeder section, prioritizing section 3, then section 1, and finally section 2. While the trend of metrics ϕ and Ω are similar, metric ϕ relates the overheating risk with the power required to avoid this risk, which provides a sensitivity for planners and decision-makers on where to invest to create the highest impact reducing the overheating risk.

Now, let us consider that a DER is installed, which is capable of supplying energy to feeder section 3 for two hours during the rotating outage. The overheating experience at each building type for an outage duration of 2 hours is displayed in Table I. The value of metrics Ψ , Π , ϕ , and Ω in section 3 in this scenario are 22.52 °C-h, 19.34 °C-h, 0.691 °C-h/kWh, and 4.836 °C-h/consumer, respectively. Adding extra capacity to feeder section 3 significantly reduces the outage-induced overheating for high-risk consumers which also considerably impacts the section total overheating perceived. Metric ϕ is comparable between feeder sections, indicating that the sections are equitably affected by the heat wave when relating the overheating to the energy curtailed. From this point, adding DER capacity in section 3

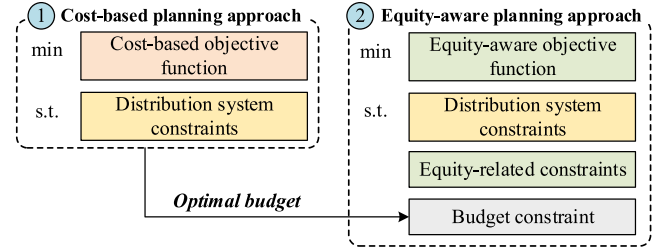


Fig. 4. The planning approach for an equity-aware distributed energy resource siting and sizing.

to continue reducing the overheating risk would lead to an inequitable operation scenario. Instead, installing upgrades in other sections helps tackle the overheating of other high-risk households distributed throughout the system.

Since metric ϕ provides a link between the overheating risk and the energy curtailed, the metric provides insights on where to upgrade to cause a significant impact on the overheating risk perceived by the community, which changes as the consumer power interruption time decreases with the extra resources. Therefore, metric ϕ is selected to be integrated into the DER planning to mitigate the overheating risk during heat waves. Metric ϕ is proposed to be applied to groups of residential consumers only, whose electricity is predominantly consumed for cooling system operation. The proposed approach, which relates the overheating risk metric ϕ with the energy curtailed, has the potential to be extended to other extreme temperature weather events, such as cold waves, which may lead to power outages and expose communities to unsafe low temperatures. For this purpose, a metric that considers household vulnerability to cold exposure should be integrated into the formulation.

III. THE PROPOSED EQUITY-AWARE PLANNING METHOD

Heat waves have a widespread effect on the power grid operation and the population exposed to extreme heat. First, heat waves lead to an emergency operational state that enforces the application of rotating outages to avoid large-scale blackouts. Second, heat waves expose consumers to unsafe temperatures, leading to the risk of overheating, which is aggravated when rotating outages are applied. This double impact can be mitigated by investing in DER upgrades that provide electricity during rotating outages in the presence of extreme heat. Power utilities plan on DER upgrades—solar generation or battery ES systems—based on a cost-centered approaches to minimize the cost of investment and operation, as well as the outage and cost of electricity from the transmission system. For the purpose of shifting into equity-aware approach to mitigate the overheating risk disparity, a framework is proposed to optimize an overheating risk metric while using the same economic budget defined in a least-cost approach. Therefore, the objective of this methodology is not to optimize the level of investment, but instead to find a more equitable DER deployment solution for a particular investment budget. The two steps of the methodology are summarized in Fig. 4, where a cost-based planning model is solved first to determine the optimal investment to be made

on the power distribution system, which is then utilized as the budget limit in the equity-aware planning approach.

The DER allocation resulting from the application of this model is expected to reduce the overheating risk when the community is exposed to a rotating outage during a heat wave event. Since rotating outages are decided at a transmission level, the proposed planning model considers the rotating outage schedule as an input for the optimization. Other events occurring during the utility rotating outage are out of the scope of this paper. Regarding the DER operation during the rotating outage, it is assumed that added DERs are capable of sustaining the operation within the section for a certain period given the capacity installed, starting at the beginning of the rotating outage. Once the capacity runs out, the nodes in the section are de-energized and the overheating risk is determined based on the remaining time under power outage until the power restoration. The model formulations are detailed below.

A. Objective Function of Cost-Based Planning Approach

In a cost-based planning approach, the objective is to decide on DER siting and sizing to minimize the annualized fixed and variable investment in solar generation and ES systems in a power distribution system, along with the DER operation cost, the cost of the electricity procured from the transmission system and the cost of the energy curtailment during the rotating outage. The objective function of a cost-based planning model is given in (10):

$$\begin{aligned} \min \sum_{i \in \mathcal{B}'} [A_{pv} c f^g \alpha_i^g + A_{es} c f^e \alpha_i^e] \\ + \sum_{i \in \mathcal{B}'} (A_{pv} c v^g p_i^{g, \text{new}} + A_{es} (c v^e p_i^{e, \text{new}} + c v^s E_i^{\text{new}})) \\ + \sum_{i \in \mathcal{B}'} (o f^g p_i^{g, \text{new}} + (o f^e p_i^{e, \text{new}} + o f^s E_i^{\text{new}})) \\ + \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{B}'} (o v^g p_{i,t}^g + o v^e (p_{i,t}^c + p_{i,t}^d)) \\ + \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{B}} p_{i,t}^{\text{grid}} \lambda_t^{\text{grid}} + \frac{1}{N} \sum_{t \in \mathcal{T}^*} \sum_{i \in \mathcal{B}} c_i (1 - e_{i,t}) p_{i,t}^l \Delta t. \quad (10) \end{aligned}$$

The first term in (10) refers to the annualized fixed investment, where α_i^g, α_i^e are binary variables that decide if a solar unit or ES system is installed at node i , respectively, $c f^g, c f^e$ are respectively the solar unit and ES system fixed installation costs, and $\mathcal{B}'$ is the set of candidate nodes where upgrades can be installed. The terms A_{pv} and A_{es} are added to annualize the investment costs associated to solar units and ES systems, respectively, calculated as a function of their discount rate r_{pv}, r_{es} and the investment time T_{pv}, T_{es} , as $A_{pv} = r_{pv} / (1 - (1 + r_{pv})^{-T_{pv}})$, and $A_{es} = r_{es} / (1 - (1 + r_{es})^{-T_{es}})$. The second term in (10) corresponds to the annualized variable investment which depends on the installed capacity, where $p_i^{g, \text{new}}$ is the added solar capacity at unit price $c v^g$, $p_i^{e, \text{new}}$ is the added ES rated charge/discharge with unit price $c v^e$, and E_i^{new} is the added ES capacity with unit price $c v^s$. The third term refers to the fixed operation costs of the installed DERs, expressed as a function of the added capacity, where

$o f^g, o f^e, o f^s$ are the annual fixed maintenance cost per solar kW, energy storage kW, and energy storage kWh, respectively. The fourth term in (10) corresponds to the variable operation cost calculated as a function of the power output of solar and energy storage units, where $p_{i,t}^g$ is the solar power injected at node i during time t with unit cost per hour $o v^g$, $p_{i,t}^c, p_{i,t}^d$ are the charging/discharging power to/from the ES system i at time t with a unit cost per hour $o v^e$, and $\mathcal{T}$ is the set of time steps for one year. The fifth term refers to the cost of the electricity procured from the bulk transmission system, where $p_{i,t}^{\text{grid}}$ is the power injected from the bulk transmission system to node i , time t , λ_t^{grid} is the locational marginal price at time t , and $\mathcal{B}$ is the set of power distribution system nodes. Finally, the last term corresponds to the cost of the energy curtailment during power outages, where c_i is the value of lost load at node i , $p_{i,t}^l$ is the active load connected at node i , time t , and $e_{i,t}$ is a binary variable that indicates if a node is energized during a rotating outage. The last term in objective function (10) considers the outage cost for all rotating outages that may occur during extreme heat, where the duration is represented by the time steps in set $\mathcal{T}^*$. The factor $1/N$ is included as a weight to represent the number of rotating power outages during heat waves per year. It is important to note that this component captures the generic cost each of kWh not served during a heat wave, without any social measure for the impact of power interruptions on vulnerable groups.

B. Objective Function of Equity-Aware Planning Approach

Considering that communities are inequitably burdened by outages during heat waves, the equity-aware planning approach can be formulated where the siting and sizing of DERs is defined by minimizing the overheating risk metric subject to a budget constraint. For this model, the objective function is formulated as:

$$\min \quad \bar{\Phi}, \quad (11)$$

where $\bar{\Phi}$ is the maximum value of the outage-induced overheating risk per kWh among the power distribution system sections $\phi_{a,\hat{t},D}$ calculated as a function of energization status of the distribution section, i.e., $\bar{\Phi} = \max\{\phi_{a,\hat{t},D}\}, \forall a$.

C. Budget and Equity-Related Constraints

1) *Integrating the Overheating Risk Metric Estimation:* DER inclusion into the power distribution system as part of an equity-aware planning implies that additional energy resources reduce the total time that groups of consumers are de-energized during the rotating outage. For this purpose, the overheating calculation becomes dependent on the energization status of the distribution section during the rotating outage. To facilitate the implementation of time-dependent metrics into the optimization, the overheating risk per kWh ϕ is expressed in piecewise linear fashion as a function of the distribution section energization status, considering increments in the overheating risk per each period that the system remains de-energized in the evaluation

horizon $\mathcal{T}'$. The metric is reformulated as:

$$\tilde{\phi}_{a,\hat{t},D} = \sum_{t \in \mathcal{T}'} (1 - \hat{e}_{a,t}) \left(\phi_{a,t,(\bar{t}-t)} - \phi_{a,t+\Delta t,(\bar{t}-t-\Delta t)} \right), \quad (12)$$

where $\bar{t}$ is the time when power is restored, $\hat{e}_{a,t}$ is a binary variable that indicates if section a is energized at time t . The calculation of the maximum overheating risk among the distribution sections is achieved with the following constraints:

$$\bar{\Phi} \geq \tilde{\phi}_{a,\hat{t},D}, \quad \forall a. \quad (13)$$

Constraints (14) and (15) are added to meet the modeling assumptions described in the previous step. Constraint (14) ensures that the energization status of the section is equal to the energization status of each of the nodes within the section, and constraint (15) prevents switching back on once the section is de-energized during the rotating outage.

$$\hat{e}_{a,t} = e_{i,t}, \quad \forall a, \forall i \in \mathcal{B}_a, \quad (14)$$

$$e_{i,t} \leq e_{i,t-1}, \quad \forall i, \forall t \in [\hat{t}, \bar{t}]. \quad (15)$$

2) *Budget Constraint*: The investments to be done in the power distribution system are limited to the available budget for the project (16):

$$\sum_{i \in \mathcal{B}'} (cf^g \alpha_i^g + cf^e \alpha_i^e + cv^g p_i^{g,new} + cv^e p_i^{e,new} + cv^s E_i^{new}) \leq \bar{B} \quad (16)$$

where $\bar{B}$ is the maximum available budget determined by the cost-based planning approach.

D. Distribution System Constraints

1) *Grid Operation Constraints*: The active and reactive power balance is shown in (17)–(18), where $p_{i,t}^{grid}, q_{i,t}^{grid}$ are the active and reactive power injections from the bulk transmission system to node i at time t , $p_{i,t}^g, q_{i,t}^g$ are the active and reactive power supplied by the solar unit installed at node i , time t , $p_{i,t}^d, p_{i,t}^c$ are respectively the discharge and charge power flow from the energy storage system installed at node i , time i , $q_{i,t}^e$ is the reactive power supplied by the energy storage systems installed at node i , time t , $p_{i,t}^l, q_{i,t}^l$ are the aggregated active and reactive loads at node i , time t , $p_{ij,t}, q_{ij,t}$ are the active and reactive power flows from node i to j at time t , $j|i \rightarrow j$ denotes the power flowing from i to nodes j , and $j|j \rightarrow i$ denotes the flows entering node i from nodes j . The power flows are limited and subject to the status of the automated switches as given in (19)–(20), where $s_{ij,t}$ is a binary parameter denoting the status of the automated switch installed between nodes i and j , $ij \in \mathcal{S}$ where $\mathcal{S} \subset \mathcal{L}$ is the subset of line sections equipped with an automated switch. The parameter $s_{ij,t}$ is equal to zero for $t \in \mathcal{T}'$ and equal to one, otherwise. The switch status also constrains the energization status of nodes that are connected through a line, so that nodes connected through a line or a closed switch must share the same energization status (21). The voltage drop between nodes is calculated based on (22) and (23), where $v_{i,t}$ is the squared voltage at node i , time t , r_{ij}, x_{ij} are respectively the resistance and reactance of line ij , and M is large real positive

number. The last term in these constraints is added to uncouple the voltages between two nodes if the switch connecting them is open. The voltage limits are given by (24), where $\bar{v}, \underline{v}$ are the squared upper and lower voltage limits, respectively:

$$\begin{aligned} p_{i,t}^{grid} + p_{i,t}^g + p_{i,t}^d - p_{i,t}^c - p_{i,t}^l e_{i,t} \\ = \sum_{j|i \rightarrow j} p_{ij,t} - \sum_{j|j \rightarrow i} p_{ji,t}, \quad \forall i, \forall t, \end{aligned} \quad (17)$$

$$\begin{aligned} q_{i,t}^{grid} + q_{i,t}^g + q_{i,t}^e - q_{i,t}^l e_{i,t} \\ = \sum_{j|i \rightarrow j} q_{ij,t} - \sum_{j|j \rightarrow i} q_{ji,t}, \quad \forall i, \forall t, \end{aligned} \quad (18)$$

$$-\bar{p}_{ij} s_{ij,t} \leq p_{ij,t} \leq \bar{p}_{ij} s_{ij,t}, \quad \forall ij, \forall t, \quad (19)$$

$$-\bar{q}_{ij} s_{ij,t} \leq q_{ij,t} \leq \bar{q}_{ij} s_{ij,t}, \quad \forall ij, \forall t, \quad (20)$$

$$|e_i - e_j| \leq 1 - s_{ij,t}, \quad \forall ij, \forall t, \quad (21)$$

$$v_{j,t} \leq v_{i,t} - 2(r_{i,j} p_{ij,t} + x_{i,j} q_{ij,t}) + M(1 - s_{ij,t}), \quad \forall ij, \forall t, \quad (22)$$

$$v_{j,t} \geq v_{i,t} - 2(r_{i,j} p_{ij,t} + x_{i,j} q_{ij,t}) - M(1 - s_{ij,t}), \quad \forall ij, \forall t, \quad (23)$$

$$\underline{v} \leq v_{i,t} \leq \bar{v}, \quad \forall i, \forall t. \quad (24)$$

2) *Solar and Storage Constraints*: The DER siting and sizing constraints are defined by (25)–(34). The DER siting is controlled by integer variables α_i^g and α_i^e , which define if a solar unit or an energy storage system is installed at node i , respectively. The solar unit sizing is defined in (25), where $\bar{p}^g$ is the maximum allowed solar capacity per node. Similarly, the battery output and capacity of the added ES system is defined in (26)–(27), where $\bar{p}^e$ and $\bar{E}$ are the maximum power output and capacity for the ES system installed at node i , respectively. The solar power output is bounded as given in (28)–(29), where r_t^{solar} is the normalized solar radiation at time t , and $\bar{q}_i^g$ is the maximum reactive power to be supplied by the inverter. The energy stored in the ES system is determined by the state equation (30), where $E_{i,t}$ is the energy stored in the ES system i at time t , η_i^c, η_i^d are the charging and discharging efficiencies, respectively, and Δt is the time step between samples. The charge and discharge limits are respectively given by (31)–(32), where $\beta_{i,t}$ is a binary variable that enables the charging operation. Finally, the available energy is limited in (33), where $\underline{E}_i$ is the minimum capacity required in the battery:

$$0 \leq p_i^{g,new} \leq \bar{p}^g \alpha_i^g, \quad (25)$$

$$0 \leq p_i^{e,new} \leq \bar{p}^e \alpha_i^e, \quad (26)$$

$$\underline{E}_i \leq E_i^{new} \leq \bar{E} \alpha_i^e, \quad (27)$$

$$0 \leq p_{i,t}^g \leq p_i^{g,new} r_t^{\text{solar}}, \quad (28)$$

$$-\bar{q}_i^g \leq q_{i,t}^g \leq \bar{q}_i^g, \quad (29)$$

$$E_{i,t} = E_{i,t-1} + (\eta_i^c p_{i,t}^c - p_{i,t}^d / \eta_i^d) \Delta t, \quad \forall t \in \mathcal{T} \setminus \{1\} \quad (30)$$

$$0 \leq p_{i,t}^c \leq p_i^{e,new} \beta_{i,t}, \quad (31)$$

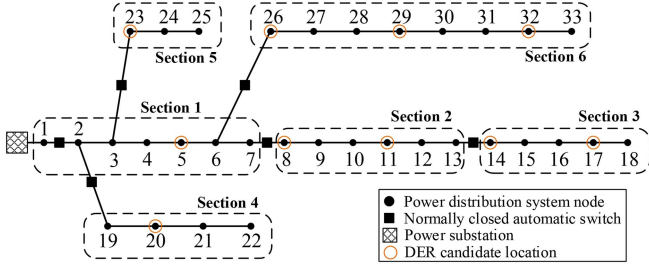


Fig. 5. IEEE 33-node system for testing.

$$0 \leq p_{i,t}^d \leq p_i^{e,\text{new}}(1 - \beta_{i,t}), \quad (32)$$

$$E_i \leq E_{i,t} \leq E_i^{\text{new}}, \quad (33)$$

$$\alpha_i^g = 0, \alpha_i^e = 0, \quad \forall i \notin \mathcal{B}'. \quad (34)$$

3) *Linearizing charge/discharge Constraints*: Constraints (31)–(32) feature the product of continuous variable $p_i^{e,\text{new}}$ and binary variable $\beta_{i,t}$, leading to a mixed-integer non-linear optimization model. In this model, the McCormick envelope is included to replace the nonlinear constraints. Let $z_{i,t}$ be the result of the product of $p_i^{e,\text{new}}$ and $\beta_{i,t}$. Given that $p_i^{e,\text{new}} \geq 0$ and that the maximum value that $p_i^{e,\text{new}}$ can assume is $\bar{p}_e$, the resulting constraints are given by:

$$0 \leq p_{i,t}^c \leq z_{i,t}, \quad (35)$$

$$0 \leq p_{i,t}^d \leq p_i^{e,\text{new}} - z_{i,t}, \quad (36)$$

$$z_{i,t} \leq \beta_{i,t} \bar{p}_e, \quad (37)$$

$$z_{i,t} \geq 0, \quad (38)$$

$$z_{i,t} \leq p_i^{e,\text{new}}, \quad (39)$$

$$z_{i,t} \geq p_i^{e,\text{new}} - (1 - \beta_{i,t}) \bar{p}_e, \quad (40)$$

where $z_{i,t}$ is equal to zero if $\beta_{i,t} = 0$, and is equal to $p_i^{e,\text{new}}$ if $\beta_{i,t} = 1$.

IV. CASE STUDY

The benefit of an equity-aware criterion on the DER siting and sizing to mitigate the outage-induced overheating risk and its comparison with a traditional cost-based planning is demonstrated in this section. The models are evaluated for DER planning on a modified IEEE 33-node test distribution system shown in Fig. 5 in the displayed candidate locations, subject to two days of a heat wave and 4-hour long rotating outages starting at hour 16 on each heat wave day.

Data used for planning considers one year planning horizon where one weekday and one weekend day of each month obtained from CIMIS for 2022 [36] are utilized to run the model. In addition, hourly data of two heat wave days with outdoor air temperature of Durham Station from September 5th, 2022 at 00:00 h to September 7th, 2022 at 00:00 h are included. Locational marginal prices (LMPs) for the same time stamps are obtained from California Independent System Operator (CAISO) [37]. Solar radiation, load profiles, and LMPs

TABLE III
RESIDENTIAL BUILDING ENVELOPE PARAMETERS

Household type	High	Medium	Mid-to-low	Low
Thermal resist. (kWh/°C)	1.75	2.25	2.75	3.25
Thermal capac. (°C/kW)	1.75	2.0	2.25	3.0
Coef. of performance	1.7	1.7	2.5	2.5
Max. power input (kW)	4.0	4.0	4.0	4.0

TABLE IV
NUMBER OF BUILDING TYPES IN SECTIONS OF THE DISTRIBUTION SYSTEM

Sect.	High	Medium	Mid-to-low	Low	Total	% type 1
1	8	6	21	62	97	8.2
2	63	31	12	5	111	56.8
3	13	7	8	28	56	23.2
4	0	0	2	7	9	0
5	5	1	0	0	6	83.3
6	12	8	10	3	33	36.4
Total	101	53	53	105		

TABLE V
SOLAR AND ENERGY STORAGE INVESTMENT AND OPERATION COSTS

	Investment		Operation	
	Fixed (\$/unit)	Variable (\$/kW, \$/kWh)	Fixed (\$/kW)	Variable (\$/kW)
PV	2400	2400	120	0.02
ES	563	563	11.26	0.02

for the representative days are provided in [38]. The $\phi_{a,t,\bar{t}-t}$ coefficients in (12) are calculated by simulating the indoor air temperature for different outage duration in steps of 15 minutes. Four residential building types are considered in the studies, with parameters given in Table III, which are derived from reference values in [39] to represent different sensitivity to temperature during the power outage. The number of these archetypes in the power distribution sections is summarized in Table IV. The investment and operation costs are shown in Table V. The indoor temperature setpoint is fixed to 26 °C, considering a condition where all consumers have increased the temperature to reduce their cooling system energy consumption while maintaining the indoor air temperature at a safe value. The simulations are run on a PC with Dual AMD Rome 7742, 128 cores, 3.6 GHz CPU.

A. Comparison Between Cost-Based and Equity-Aware Planning Solutions

As explained in Section III, the cost-based optimization approach is evaluated first to determine the optimal location of DERs that minimizes the costs considered in the objective function, and then the investment cost obtained from the solution is used to set the budget constraint upper bound $\bar{B}$ in (16) for the equity-aware planning model.

The optimal budget after solving the cost-based optimization is \$3,512,318, and the resulting designs for both optimization models are shown in Fig. 6, with power output and capacities given in Table VI. In both planning approaches, the optimization leads to the installation of ES systems only with no solar generation included. As the rotating outage is expected to occur during the peak load occurring near the end of the day where solar power output is reduced, ES is preferred to ensure the

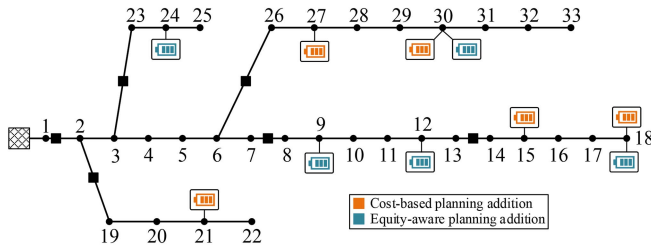


Fig. 6. Location of distributed energy resources after optimization: Cost-based planning approach (orange), proposed equity-aware planning approach (blue).

TABLE VI
DER INVESTMENT SOLUTIONS FOR THE PROPOSED MODELS

Node	Cost based		Equity aware	
	ES output (kW)	ES capacity (kWh)	ES output (kW)	ES capacity (kWh)
9	—	—	530.94	1152.59
12	—	—	675.16	1500
15	271.73	1055.42	—	—
18	394.41	1500	624.99	465.97
21	134.68	516.64	—	—
24	—	—	72.35	176.35
27	160.75	578.68	—	—
30	326.36	1289.90	457.01	571.22

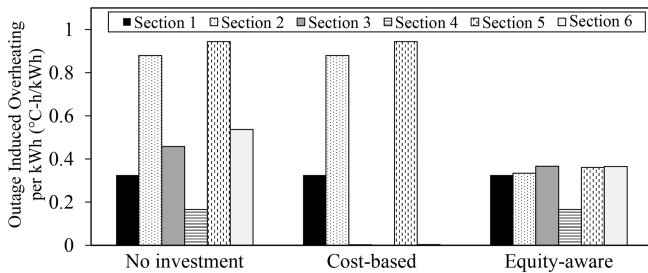


Fig. 7. Section-based outage-induced overheating risk for the proposed models.

energy availability during the power interruption. The optimal DER location is different in the two design approaches, with the exception of nodes 18 and 30 that were selected as both economic and equitable locations for DERs. The cost-based planning prioritizes the DER installation in sections where the cost of the outage can be quickly reduced, as upgrades in sections 1 and 2 lead to higher investment costs not compensated by other variables. Instead, the equity-aware planning ensures that sections with higher concentration of high overheating risk houses are prioritized in the DER distribution, e.g., sections 2 and 5.

Each planning model leads to different performance in terms of overheating risk reduction, as shown by the overheating risk per kWh for each section in Fig. 7. The cost-based planning ensures that sections where DERs are installed are fully energized during the rotating outage, leading to zero value for the metric and no change for sections 1, 2, and 5. Feeder section 2 is critical, as it contains the highest number of consumers and the highest concentration of high overheating risk consumers.

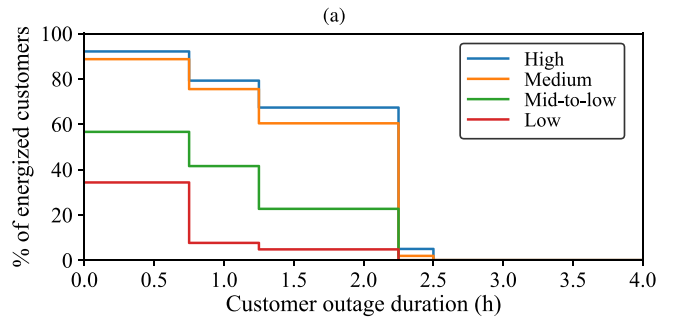
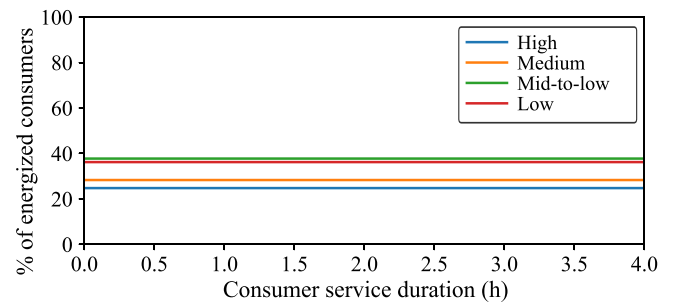


Fig. 8. Percentage of energized consumers vs consumer service duration per household type: (a) Cost-based solution. (b) Equity-aware solution.

In this case, the cost-based planning intensifies the heat disparity between the sections. On the other hand, the equity-aware planning leads to a scenario where the avoidable overheating risk is comparable among the sections. The metric is reduced more than 50% for sections 2, and 5, and moderately reduced for 3 and 6.

The impact of the DER locations on the available supply of different buildings for the cost-based and equity-aware models is shown in Fig. 8. Fig. 8(a) shows the resulting household energization for the cost-based planning, which is sustained from beginning to end throughout the rotating outage. However, nearly 40% of the consumers living in mid-to-low risk households are energized, while roughly 20% of the consumers with a high overheating risk have electricity supply available, as shown in Fig. 8(a). On the other hand, the available energy resources located based on an equity-aware approach are capable of sustaining up to 2.5 hours before the system remains de-energized until its restoration, as depicted in Fig. 8(b). Fig. 8(b) shows that more than 92% of the high risk consumer are energized for the first hour, and over 67% of the high risk consumers are energized during the second hour. Hence, the power availability is assigned based on how critical the household is in terms of overheating risk, as expected from an equity-aware standpoint. It is important to consider that the overheating risk can be reduced by delaying the de-energization starting time, as shown in Fig. 8(b), which avoids excessive increase of the metrics during the complete rotating outage.

B. Effect of Equity-Aware Planning on Operation Costs

Table VII summarizes the results of different quantities to compare the impact of cost-based and equity-aware planning

TABLE VII
VALUE COMPARISON

	Before investment	Cost-based investment	Equity-aware investment
Operation cost (\$)	1,046,560	944,137	1,050,270
Outage cost (\$)	2,723,540	1,839,160	2,049,410
Maintenance cost (\$)	—	37,902.17	25,898.33
Equity metric	0.9442	0.9442	0.3664

decisions. Including DER reduces the outage cost for both scenarios, 32.47% less for the cost-based investment and 24.75% less for the equity-aware investment. Cost-based investment prioritizes areas with larger loads where outage cost is higher, unlike equity-aware investment, which prioritizes areas with higher overheating risk. Regarding the operation and maintenance, the decreased operation cost combined with the higher maintenance cost for the cost-based investment displays an extended use of the installed DERs throughout the year to reduce the cost of energy purchased from the bulk transmission system. On the other hand, the use of DERs on the power distribution system after the equity-aware investment is prioritized during the outage periods to reduce the overheating risk inequity; however, it is worth noting that other costs are not modeled in the objective function; hence these are not optimized.

C. Overheating Disparity Sensitivity to Available Budget

In the previous section, the effect of an equity-aware planning criterion on the siting and sizing of DERs was presented based on a budget constraint defined by a cost-based planning method. In this section, the effect of the available budget on the DER allocation is discussed. The equity-aware model is evaluated for different budgets around the optimal value determined in Section IV-A. The purpose is to assess how the model prioritizes the power distribution system sections and how different budget levels mitigate overheating risk disparities across the feeder. The obtained section metrics for the considered budget scenarios are shown in Fig. 9(a). The section-based overheating risk metrics for the scenario with no investment are also included as a reference. The percent reduction in the metric value from the no-investment scenario with each budget addition is shown in Fig. 9(b).

As more budget becomes available for investment, the DER allocation progressively reduces the overheating risk metric of each area by reducing the amount of time that a section is de-energized, starting from the sections with the highest overheating risk per kWh values in the first budget step, i.e., feeder sections 2 and 5. Further increments of budget allow addressing other sections, such as number 6, where the overheating is also high. After the optimal budget is reached, further improvement can be achieved in terms of heat disparity reduction. It is possible to see that, for each budget increase, the section with worst overheating risk is improved. Higher values of the budget not only lead to less risk but also less disparity among the sections, representing a more equitable scenario related to how benefits and burdens are distributed throughout the system and how community is affected.

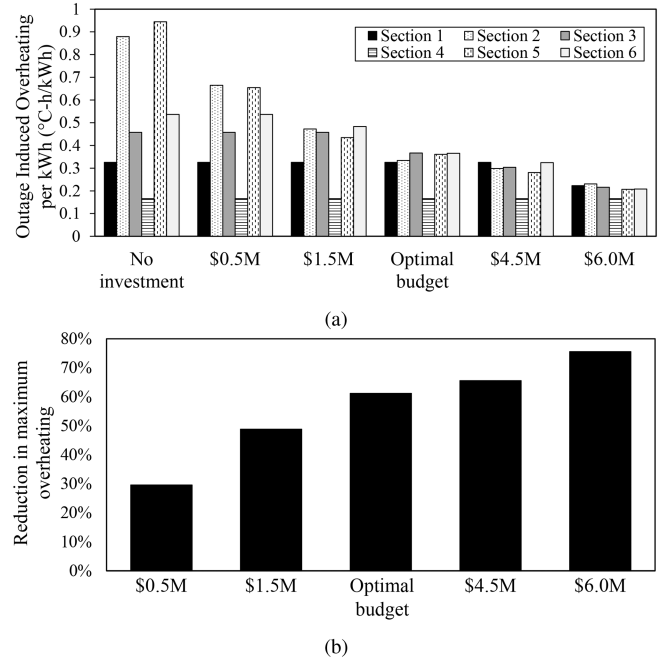


Fig. 9. (a) Effect of the budget on the equity-aware allocation of distributed energy resources. (b) Percentage reduction in the maximum value of the metric at the no-investment state with the budget increase.

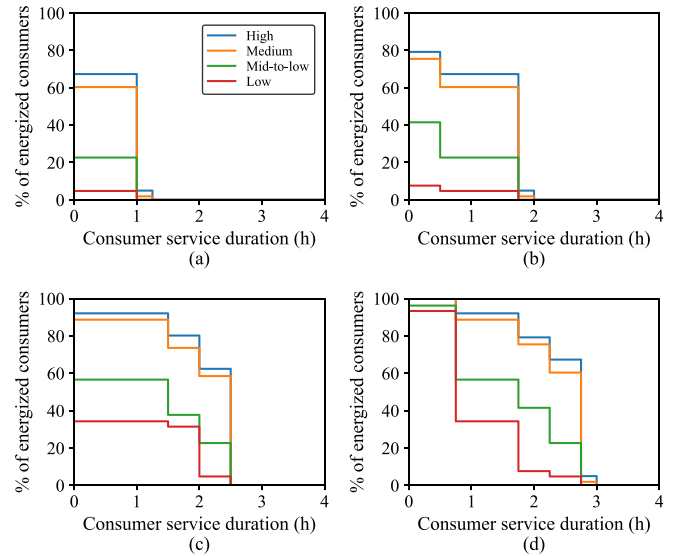


Fig. 10. Effect of the budget on the percentage of energized consumer per outage time for different budgets: (a) \$0.5 M, (b) \$1.5 M, (c) \$4.5 M, and (d) \$6.0 M.

The effect of the available budget of the energization periods per household type is shown in Fig. 10. Based on Fig. 10, the percentage of consumers supplied and the energization duration is increased as more budget for planning is available. Fig. 10(d) displays 100% energization for consumers in high risk households during the first hour, and available energization until nearly 3 hours into the rotating outage. As previously explained, the section overheating risk increases as the section

TABLE VIII
MODEL SCALABILITY

	33-node	123-node
No. candidate solutions	10	23
No. sections	6	9
No. buildings	312	588
Solution time (s)	78.07	1220.58

remains de-energized during the rotating outage. While ideally sustaining the electricity supply for the total duration of the rotating outage leads to the minimum overheating risk for the community, reductions in the de-energization time can significantly diminish the overheating risk, considering the non-linear relationship between overheating risk and the consumer outage duration.

D. Model Scalability Assessment

The proposed model is evaluated on a IEEE 123-node test system, which is split in 9 zones. The data for this system and the household distribution throughout the test systems can be found in [40]. The average time to run the equity-aware optimization for the IEEE 123-node and IEEE 33-node considering ten different budget cases, along with another parameters, is shown in Table VIII. The added nodes between cases also increases the number of decision variables in the system dependent on the number of nodes, leading to a larger average solution time comparing IEEE 33-node and 123-node. The complexity of modeling the grid elements and nodes has a direct impact on the performance of the proposed model. However, it is important to highlight that overall planning problems are complex and, in contrast to operation problems that require accurate results in short periods, planning problems are solved offline with solutions expected to be obtained within days. Despite the expected time increase with a larger system, the model shows capability to converge to a solution and is promising for implementation in larger power distribution systems.

V. CONCLUSION

Climate-change-driven extreme heat waves challenge the operation and planning of power distribution systems, compromising their resilience and exposing communities to an inequitable overheating risk. This paper introduced a planning framework to mitigate the overheating risk of communities exposed to rotating outages during an extreme heat event, which allocates distributed energy resources (solar generation and battery energy storage systems) to provide additional power and delay the de-energization of consumers during a rotating power outage, leading to a reduction in heat-related risks for the community members. The results show that the proposed framework can capture the overheating risk of vulnerable populations and prioritize DER investment as a function of the utility capital distribution investment allowed by the regulator on an equity-aware basis. In addition, the resulting set of non-dominated budget-versus-overheating risk solutions can be used as an innovative instrument to support equitable regulatory processes of distribution system planning. Future work may expand considerations

of realistic power distribution systems for the planning decisions, including three-phase configurations, e.g., [41], and the evaluation of the approach on real power distribution systems, with associated community and extreme heat data. Future works also include the expansion of equity-aware planning to address the effect of rotating outage uncertainty. Further, more advanced indoor temperature models can be utilized to improve the estimation of the overheating risk conditions in each household, and thermostat data, e.g., [42], may be utilized to improve the building parameter estimation, as applied in [43].

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